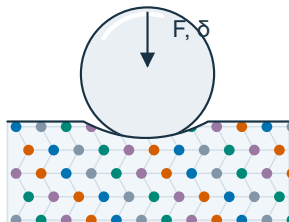


Energy-based learning of nanoindentation

AlCuNiTiZr high-entropy alloy

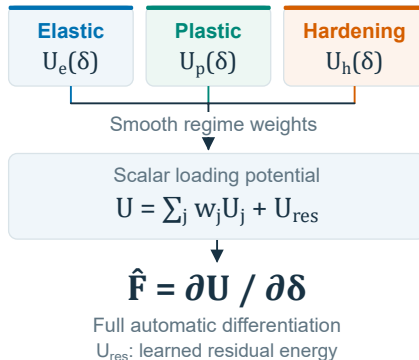
01 MD loading data



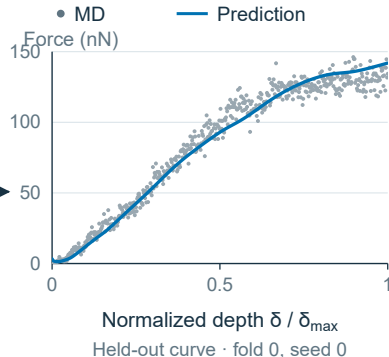
Nanoindentation · schematic

Depth δ · grain size g
 Temperature T · velocity v
 Time-ordered loading branch

02 Energy-based model



03 Unseen condition



Energy–force consistency · Interpretable regime weights

18 conditions → 14 train / 3 validation / 1 test per fold
 Leave-one-condition-out: 18 folds × 5 seeds

$R^2 = 0.957 \pm 0.040$

Mean $\pm$ SD across folds and seeds

7.07 nN

Mean MAE